host or travel?

a study on social connecton among LGBTQ New Yorkers

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Welcome to MPX NYC

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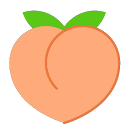
RESPND-MI Collaboration



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https://www.mpxnyc.app

1. Housing
2. Nutrition
3. Violence
4. Healthcare access
5. Employment
6. Health

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Part I

Background

1 From outbreak to epidemic

When we started, MPOX testing was not at all available. Our goal was to estimate the proportion of community members who experienced mpox-like symptoms. We also wanted to understand the structure of social and sexual networks that connect our community so that we could understand how to most effectively distribute MPOX vaccines. Within 5 months of conceiving the project, we had strated collecting data. We kept the survey open for about 1 month, obtaining about 1300 responses.The data have proven useful - we started sharing periodic analyses with the New York City Department of Health, guiding their outreach activities. Preliminary analyses were shared at the IAS international conference in Montreal, and have through several presentations and public discussions since then.As MPOX shifted from an outbreak to an epidemic, we took time to come up with the best way to make sense of these data and to convey them to different readers. This report is the result of that effort.We have prioritized the reproducibility of our study findings and our study design. Our hope is that alliances of LGBTQ activists and researchers will use these methods to answer the pressing questions of their contexts. The data and code we used to make this report are available here [HYPERLINK TO GITHUB]. We have written detailed appendices on the methods we devised to make sense of our data.As resources allow, we will publish the code for the custom-designed tool we developed to safely measure sensitive locations as well as explainer videos. Our goals is for the report to be accessible to most interested readers while maintain enough technical precision to allow others to understand, critique, and possibly replicate the study.The next section gives an overview of what scientists already know about the themes our research report touches on: the relationship between people's social networks and their health, the infectious disease dynamics of a novel outbreak, the health of

1.1 Rapid oubreak

Mpox cases in the United States (U.S.) were first detected in May 2022 and rapidly escalated, with over 10,000 cases reported by the end of July 2022 (Minhaj et al. 2022). Initial clusters were associated with large gatherings involving sex and concentrated in white, cisgender, gay, bisexual, and other men who have sex with men, with limited data available for minoritized ethno-racial and transgender communities. Over time, new data indicated that Black, Latinx, and transgender communities were also disproportionately affected (Blackburn et al. 2022).

1.2 Slow responses

Despite early warnings from activists about the lack of testing capacity in the U.S., responses from city, state, and federal governments were slow and inadequate (Krellenstein, Osmunson, and Makofane 2023). At the height of Pride month in June, only about 20 mpox tests were administered daily in New York City (NYC), in stark contrast to the 719 probable or confirmed mpox cases in NYC between May 19 to July 15, 2022 (Kyaw et al. 2022). This was a critical period for scaling prevention efforts in vulnerable communities affected by mpox, yet structural divestments led to inadequate data collection; unpredictable vaccination access and vaccine shortages (Anna Grace Lee, n.d.); and challenges in obtaining and providing treatment ("For Monkeypox Patients, Excruciating Symptoms and a Struggle for Care" 2022).

1.3 Weak systems

1.3.1 Health communication

- Limited
- Ugly
- Stigmatizing
- Inaccessible

1.3.2 Healthcare delivery

- Informal networks // Variable care if you can afford
- FQHC if you can't afford
-

1.3.3 Strategic information

- Based on biased healthcare delivery data
- Not fine-grained

1.4 A living epidemic

1.4.1 Burden

1.4.2 Transmission dynamics

1.4.3 The role of networks

2 Our response

We organized the RESPND-MI study team into three working groups, inviting community members to join regardless of academic credentials: the administration group (fundraising and Institutional Review Board applications); the technical group (questionnaire and survey instrument development); and the external affairs group (public-facing marketing, communications, and partnerships). Community partners joined as co-investigators and participated in all collective decision-making.

2.1 Administration

- First, developed a study protocol and obtained institutioal review board clearence.
- Fundraising
- Financial reporting
- Financial transactions

2.2 Technical

2.3 External affairs

The external affairs group played a crucial role in ensuring that RESPND-MI benefited from the expertise of individuals and organizations working in queer and transgender health in NYC. The team regularly consulted with stakeholders, including activists, government officials, community-based organizations, clinicians, researchers, and other community partners through the RESPND-MI's community forum.

2.3.1 RESPND-MI Community Forum and Investigator team

Starting in May 2022, we prepared a study about sexual activity and mpox symptoms among those most vulnerable to mpox. To promote consultation and collective decision making for the study, we organised an online community forum (CF). In the first CF, attendees expressed the need for a space to continually coordinate response efforts across organisations, and to address gaps not immediately met by local and federal public health agencies

CFs thus became weekly calls where participants received epidemiologic updates, discussed problems, and provided input on study design and marketing decisions. From June to September 2022, over 80 activists, community leaders, public health professionals participated in CFs. We produced policy briefs, op-eds, and educational materials; and conducted media outreach. The RST website became a clearing house for community-generated resources, as well as community-generated information on vaccination, testing, and treatment services.

Educational materials and policy recommendations from the CF were adopted by city, state, and federal PHAs. Reporters cited the RST in over 90 press clips. Survey data were used to inform NYC vaccination outreach services. (Roth et al. 2023).

The Community Forum shaped study plans, including the recruitment campaign, the design of a novel spatial data collection tool, and the development of our study questionnaire. Sustained engagement led to several concrete improvements in research approach. For example, a criticism of the initial communication approach was that it centered cisgender men to the exclusion of transgender women and transgender men. This was consistent with most communication about the social epidemiology of MPOX at the time, and most scientific reporting.

But given that many public heatlh data systems use measures for gender that are unable to identify transgender individuals, limiting attention to cisgender men would have limited our ability to understand the true extent of the mpox outbreak. We assumed that transgender people and cisgender people are part of the same social and sexual networks which would likely be affected by mpox. We invited the participant who raised that criticism to join the study team. Over time, we attempted to continually improve gender related language and iconography in the project through discussion among investigators.

3 What we know

3.1 Background

The overall goal of the RESPND-MI study is to rapidly generate actionable epidemiologic information about monkeypox transmission among gay, bisexual, and other men who have sex with men in New York City. Aim 1: Characterize the overall structure of social and sexual networks among gay, bisexual, and other men who have sex with men in New York City. Aim 2: Identify the characteristics of individuals who are central in the social and sexual networks of gay, bisexual, and other men who have sex with men in New York City. Aim 3: Assess the prevalence of self-reported symptoms monkeypox symptoms among gay, bisexual, and other men who have sex with men in New York City. Aim 4: Assess the relationships between monkeypox symptoms and HIV clinical cascade outcomes

The Rapid Epidemiologic Study of Prevalence, Networks, and Demographics of Monkeypox Infection (RESPND-MI) aims to close this knowledge gap by rapidly mapping the social and sexual networks of gay, bisexual and other men who have sex with men in New York City and providing an estimate of the prevalence of symptoms consistent with monkeypox infection. These data would help public health officials understand the scale of testing and vaccination needed, and potentially highlight the most impactful interventions to stop onward transmission of monkeypox virus. This study will not collect any individual identifiers from participants.

There are no specific treatments for monkeypox infection, and testing is not yet widely available and well-advertised. However, there are vaccines that could be deployed to slow down transmission. There are several strategies for strategic vaccine deployment to prevent onward transmission. One would be to offer voluntary vaccination to men who believe they are at risk. There may not be enough vaccine supply to do this. Another approach is ring vaccination: those who are named as recent close contacts of confirmed cases would be offered the vaccine, breaking chains of transmission. Since there is no large-scale testing program for monkeypox, however, and since contact tracing would have to be conducted among a population whose sexual activity is already stigmatized, it is not clear whether this strategy would be effective. Furthermore, ring vaccination would fail if there were already widespread community transmission.

Finally, vaccines could be deployed strategically by targeting groups and locations which are central in the overall sexual network. We propose the RESPND-MI study - a rapid online survey to map the social and sexual networks of men who have sex with men in New York City and to rapidly assess the prevalence and patterning of self-reported monkeypox symptoms. The information yielded by this study would enable public health officials to use vaccine supplies to effectively and efficiently slow down and ultimately halt the transmission of monkeypox among men who have sex with men.

3.2 Significance

The study will yield information about the overall structure of the social and sexual networks of gay, bisexual, and other men who have sex with men in New York City and it will yield information about the extent of monkeypox infection in this population. Since, to date, there are relatively few cases that have been detected, identifying the groups/areas/sub-networks where there are active chains of transmission will allow public health officials to act now to prevent a larger future epidemic. The information yielded by RESPND-MI would enable public health officials to use vaccine supplies to effectively and efficiently slow down and ultimately halt the transmission of monkeypox among men who have sex with men.

In addition, these network data will be useful to scientists and health program implementers who want to understand patterns of connection among men who have sex with men. For implementers, this will enable them to improve the targeting of healthcare services to this underserved population by disseminating health information and information about services using the social and sexual networks. For researchers, this knowledge will strengthen their understanding of the spread of sexually transmitted infections including HIV, with implications for the design and evaluation of new interventions.

Welcome to the MPX NYC Study on the social and sexual connections of LGBTQ New Yorkers. Through this website we report data we collected among LGBTQ New Yorkers in the Fall of 2022 when the first global human mpox oubreak occured.

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Part II

Methods